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Physiology: Transport of Gases in Blood & Tissue Fluids

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Objectives

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- Compare and contrast different ways, O_2 and CO_2 are transported in blood. How does Henry's Law apply to those?
- Compare and contrast structure, function and pathologic implications of adult and fetal hemoglobin, methemoglobin and sickle cell hemoglobin.
- Explain the differences between oxygen content and binding capacity.
- Explain how cooperative oxygen binding influences the sigmoidal shape of the oxygen dissociation curve. How is this shape beneficial to oxidative metabolism?
- Explain the physiological rationale behind PCO_2, pH, temperature, and 2,3-DPG being able to shift the oxygen dissociation curve (left or right). What conditions favor the 'Bohr shift' and why?
- Explain how pulse oximetry works. How does it focus on arterial and not venous or capillary blood?
- Explain the effect of CO on hemoglobin. What are the consequences and how do they matter?
- Compare and contrast the three ways CO_2 can be transported in blood. How does the Haldane effect facilitate normal metabolism?

Gas Transport in Blood O_2

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Gas	Method of Transport	Percent Carried in This Form
O_2	Physically dissolved	1.5
	Bound to hemoglobin	98.5
CO_2	Physically dissolved	10
	Bound to hemoglobin	30
	As bicarbonate (HCO_3^-)	60

In other texts:

2%

98%

- Concentration of O_2 Dissolved at 100mm Hg = **0.3 mL O_2 /100 mL blood** (100 mm Hg × 0.003 mL O_2 /100 mL blood per mm Hg; **Henry's Law**)

- 15 mL O_2 /min dissolved supplied → 250 mL O_2 /min needed at rest (13 x difference)

O_2 Bound to HbA (Adult Hemoglobin)

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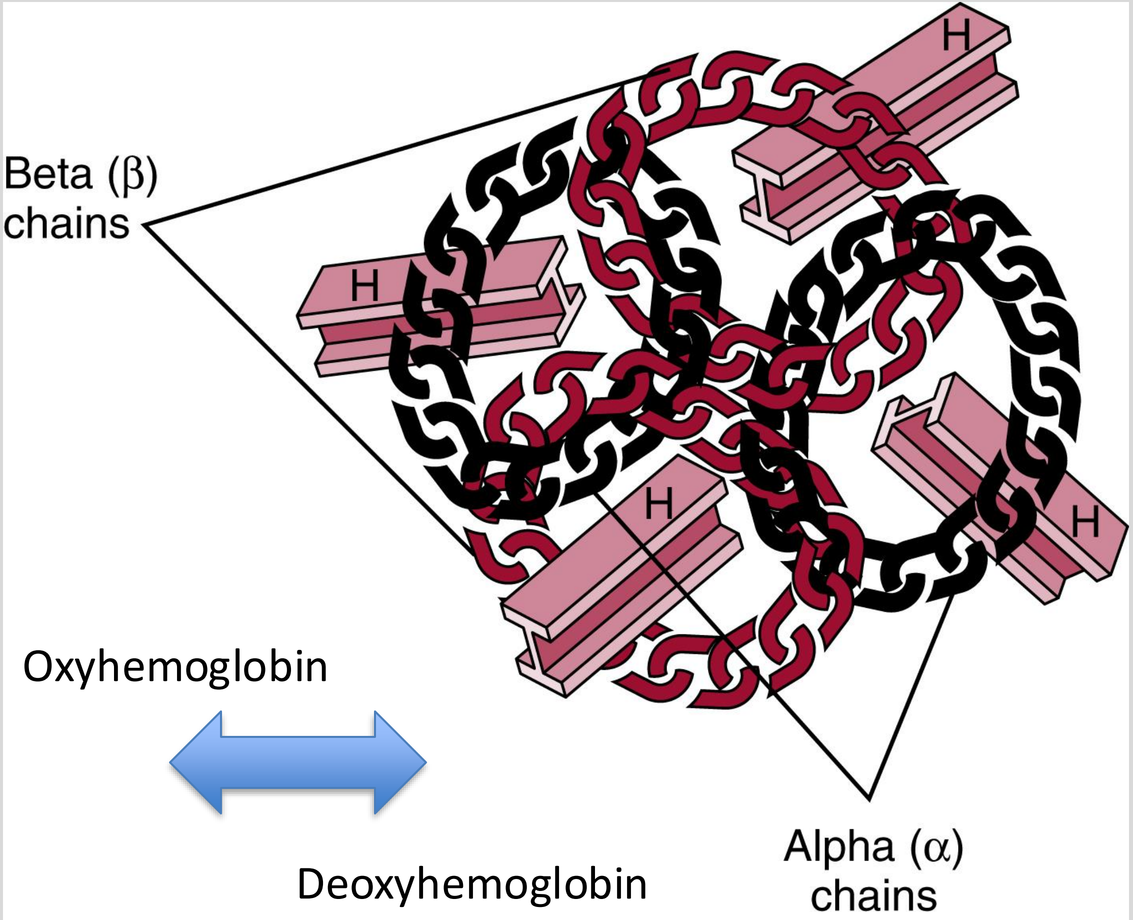

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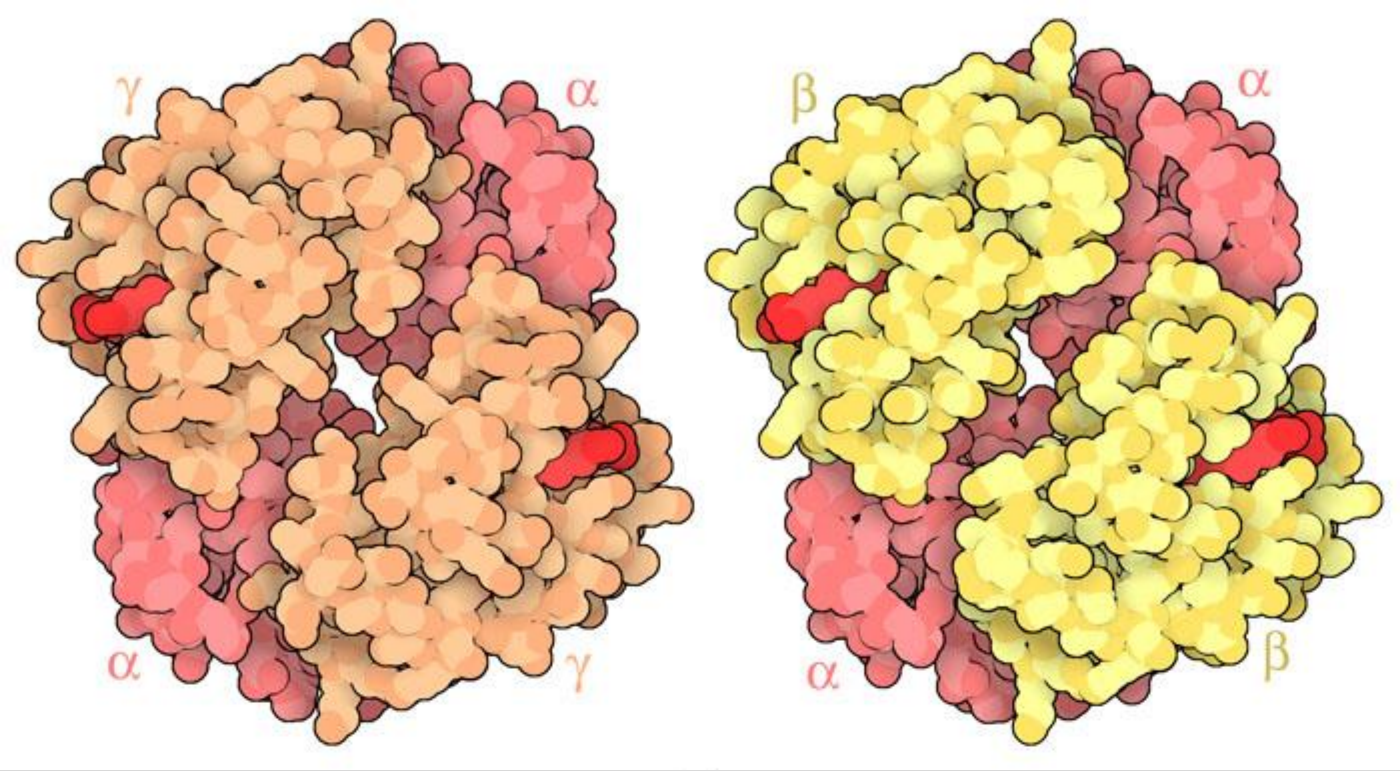
HbF: Fetal Hemoglobin



Fetal (2 alpha, 2 gamma)   Adult (2 alpha, 2 beta)

Fetal HbF:

- **Higher O_2 Affinity**
- Does **not** interact with 2,3 BPG

Methemoglobin

Unable to bind Oxygen

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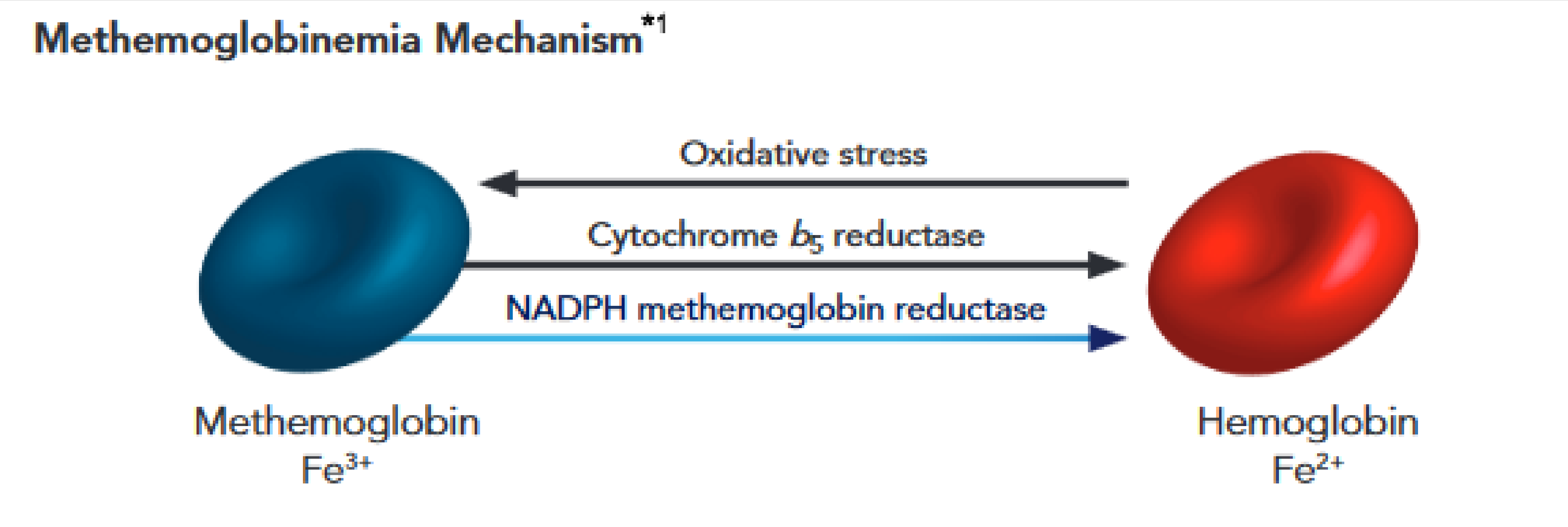
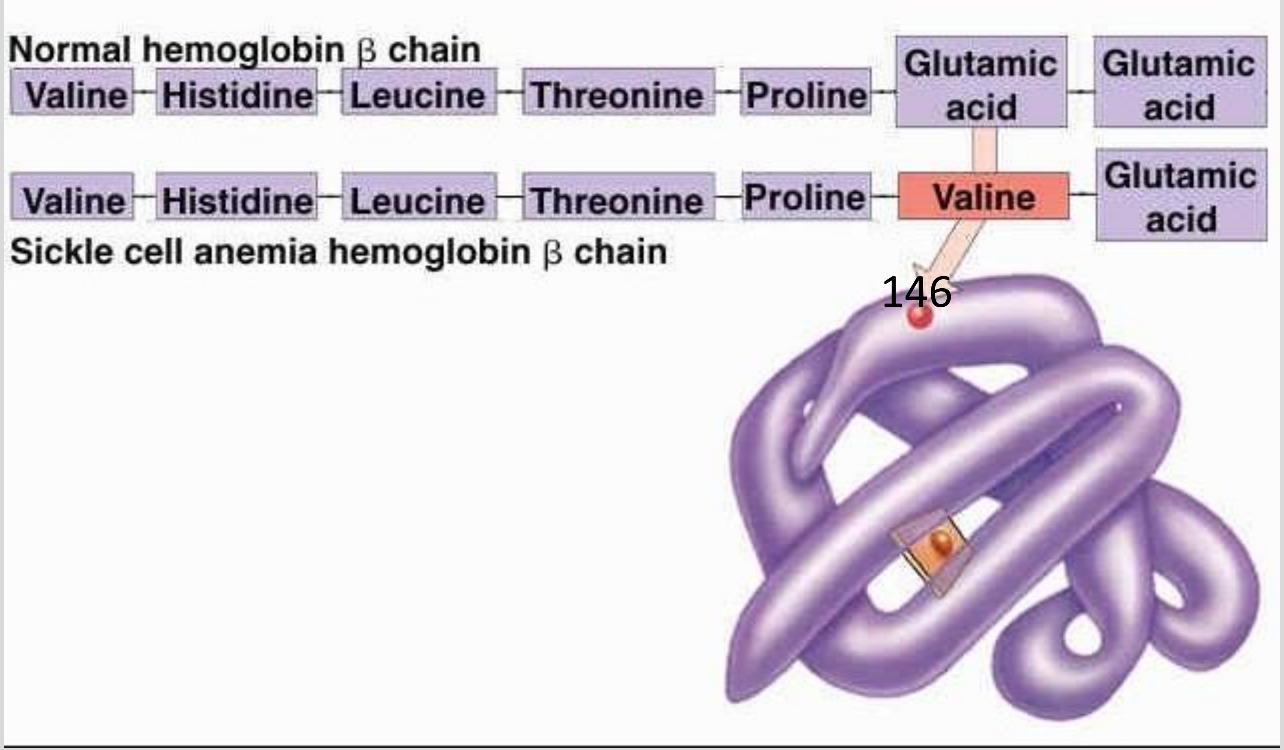


HBS: Sickle Cell Hemoglobin

Lower O2 Affinity



O_2 Binding Capacity vs. O_2 Content

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The O_2 -binding capacity is the **maximum amount of O_2** that can be **bound to hemoglobin** per volume of blood

1 g of hemoglobin A can bind 1.34 mL O_2

The normal concentration of hemoglobin A in blood is 15 g/100 mL

20.1 mL O_2 /100 mL blood maximally bound to hemoglobin
(15 g/100 mL × 1.34 mL O_2 /g hemoglobin)

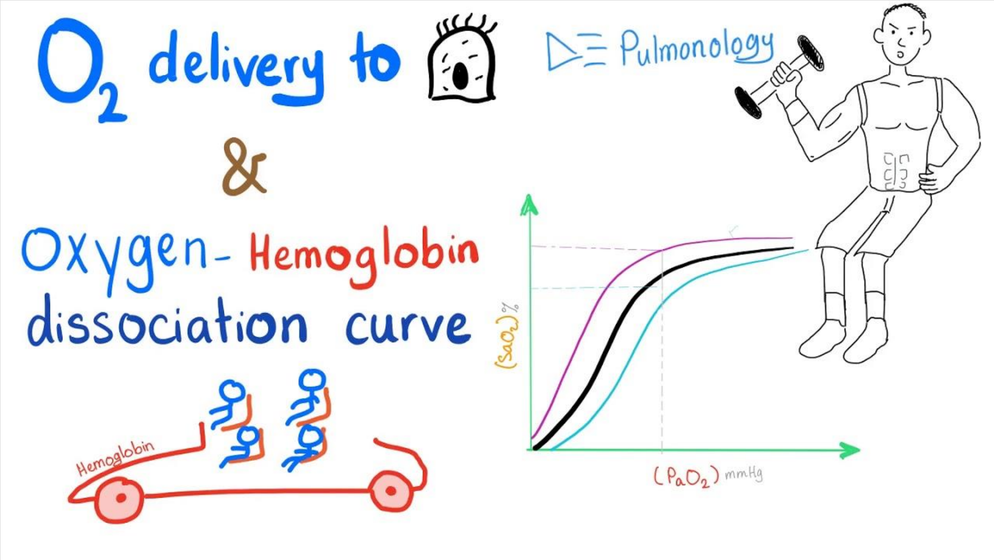
The O_2 content is the **actual amount of O_2 per volume of blood**

O_2 content = (O_2-binding capacity × %Saturation) + Dissolved O_2

O_2 Delivery to Tissues

O_2 content of blood is the sum of **dissolved O_2 (~2%)** and **O_2-hemoglobin (~98%)**

O_2 delivery = Cardiac output × O_2 content of blood
= Cardiac output × (Dissolved O_2 + O_2-hemoglobin)



O_2 – Hemoglobin Dissociation Curve

Changes in Affinity through positive Cooperativity

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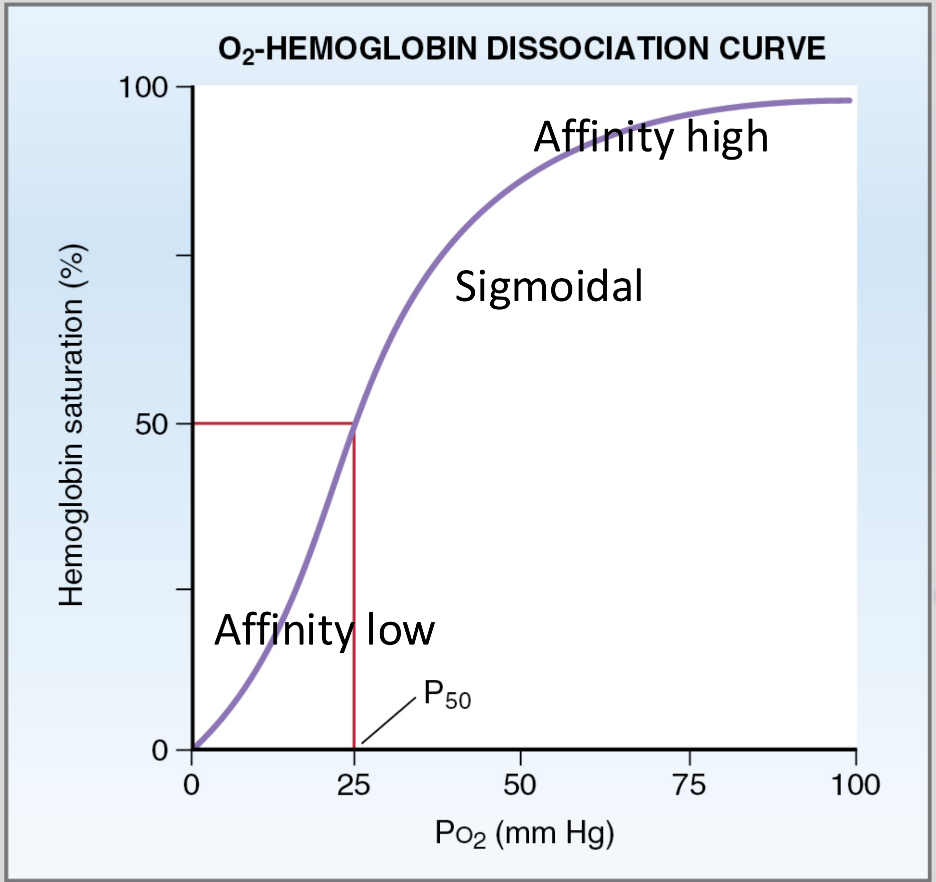
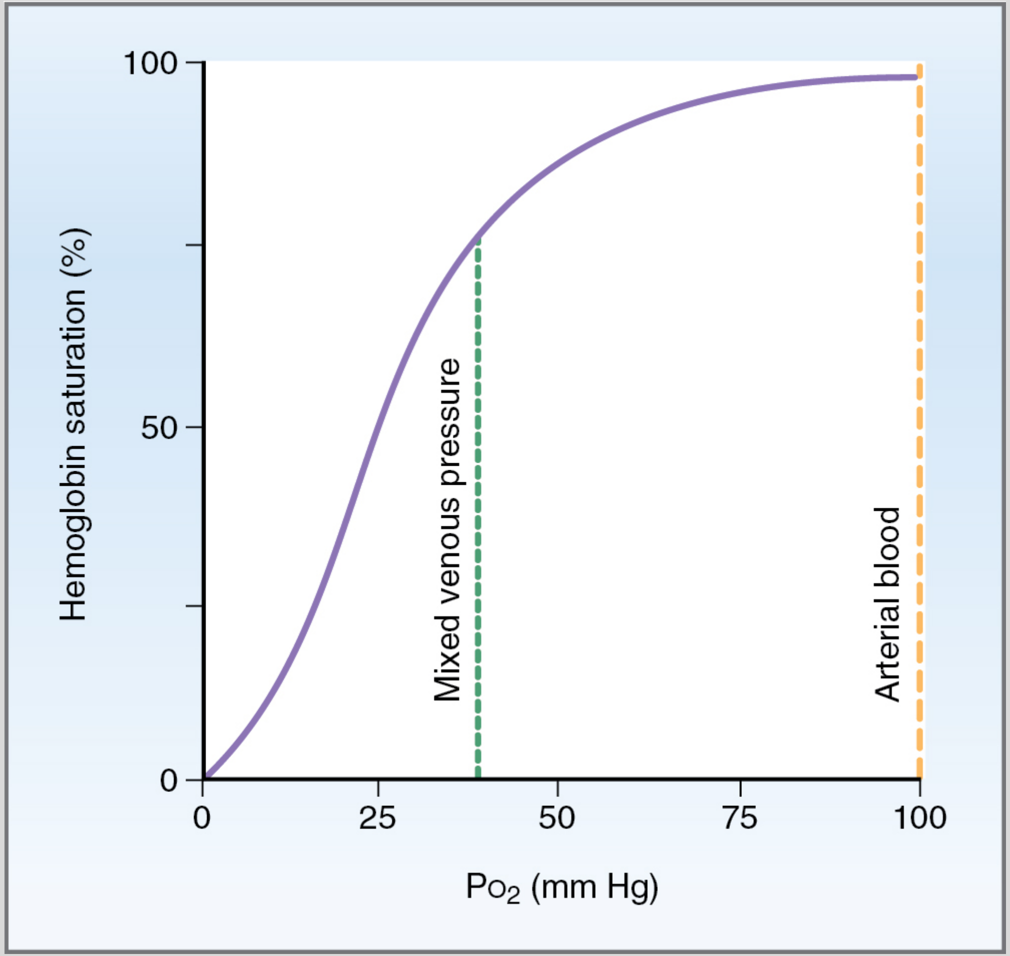


Po_2 (mm Hg)	Saturation (%)
10	25
20	35
25	50
30	60
40	75
50	85
60	90
80	96
100	98

The Po_2 that corresponds to 50% saturation of hemoglobin is called P_{50}.

Loading and Unloading of O_2



Pulse Oximetry:

- Dual-wavelength spectrophotometry oxyhemoglobin vs deoxyhemoglobin
- % saturation used to estimate arterial P_{O_2}

Changes in the O_2-Hemoglobin Dissociation Curve

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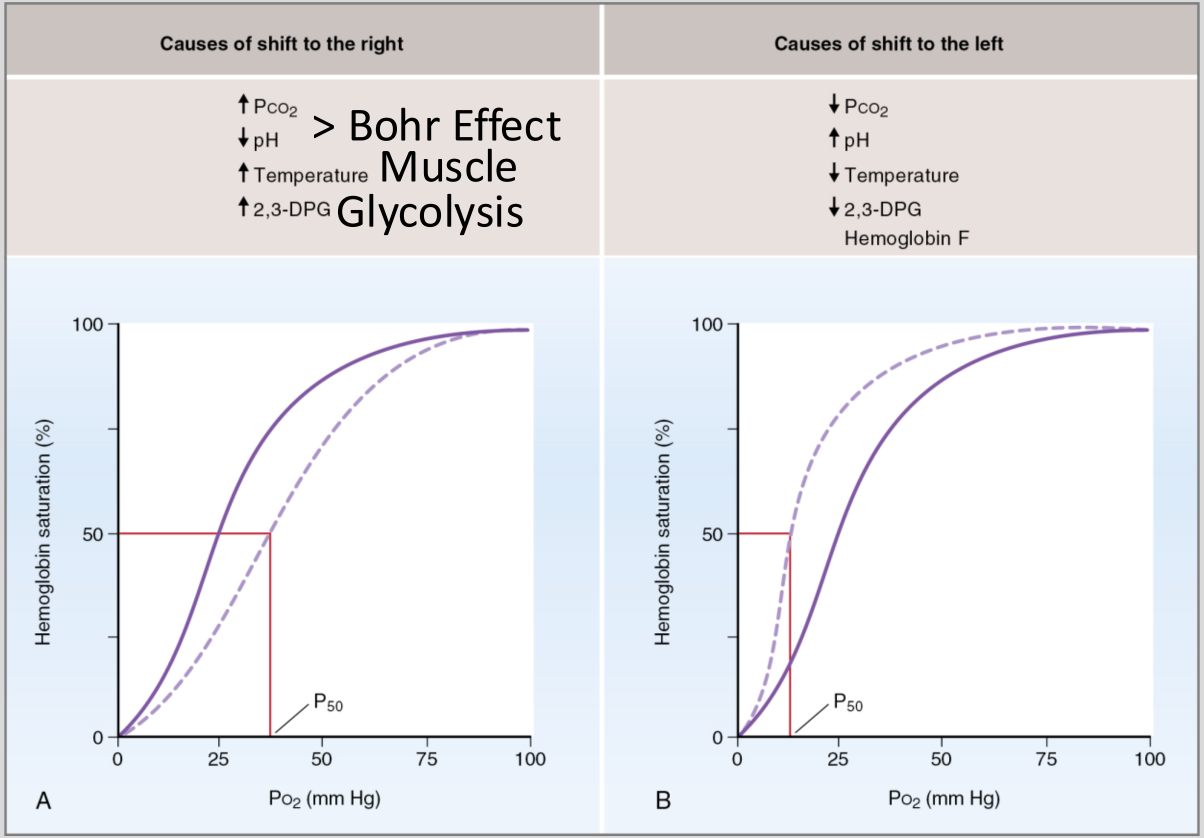

Metabolism & its Products

The Effect of CO – Carbon Monoxide

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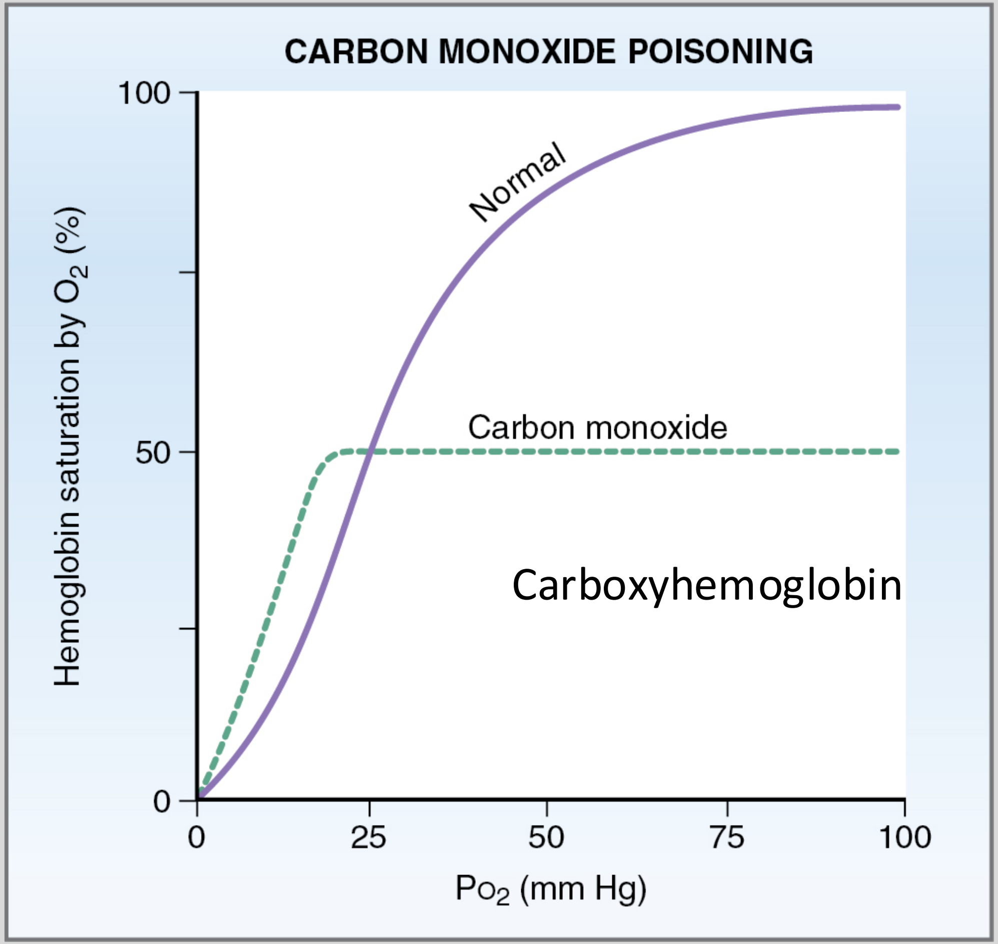
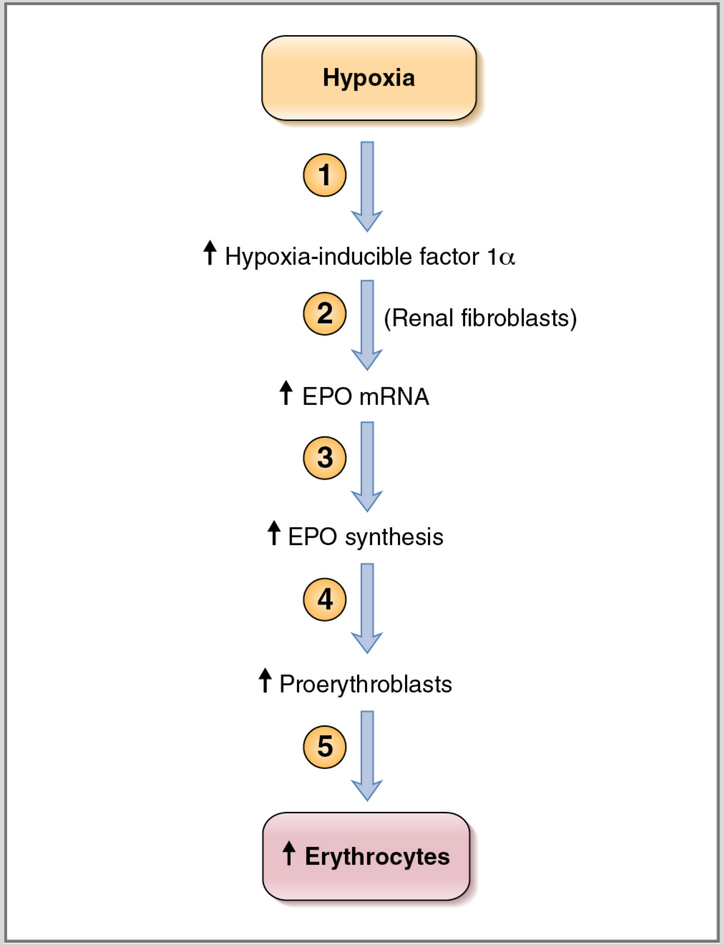


- 210-250 times higher affinity to heme than O2

Therapies:

- Regular air: ~5h
- Air saturated with O2: ~1.5h
- Hyperbaric chamber (3ATM) ~20min

Erythropoietin EPO – a Glycoprotein



Kidneys (to a much lesser degree: Liver)

Kidneys can distinguish between decreased blood flow and decreased O_2 delivery and **decreased O_2 content of arterial blood**

CO_2 Dissolved in Blood (Free)

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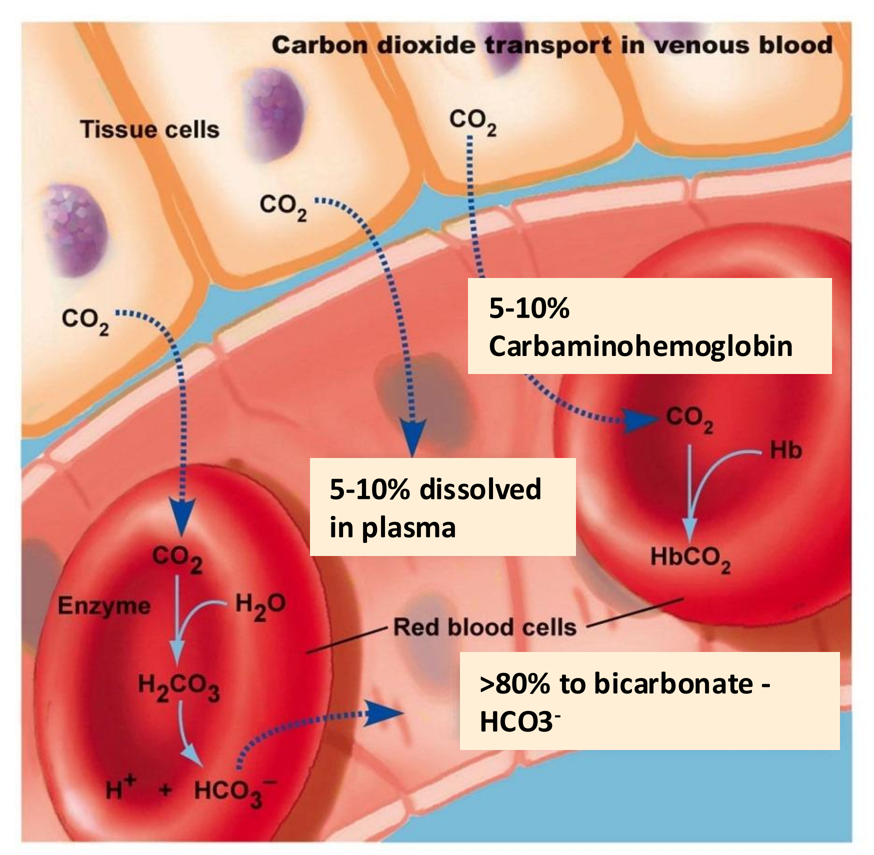

Percentages differ somewhat between texts. Stick with the values here!

2.8 mL CO_2 /100 mL blood (40 mm Hg × 0.07 mL CO_2 /100 mL blood per mm Hg)

→ **that is ~5%**

Deoxyhemoglobin binds CO_2 More Readily

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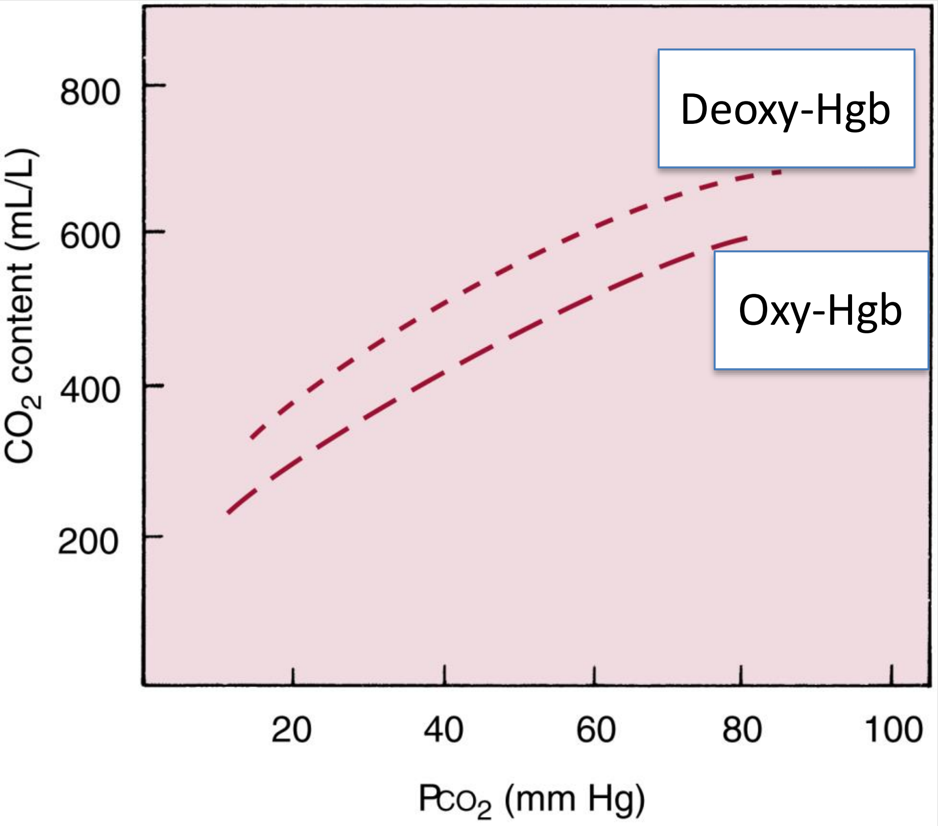

Haldane Effect:
Facilitates CO_2 uptake by deoxyhemoglobin

Bicarbonate HCO_3^-

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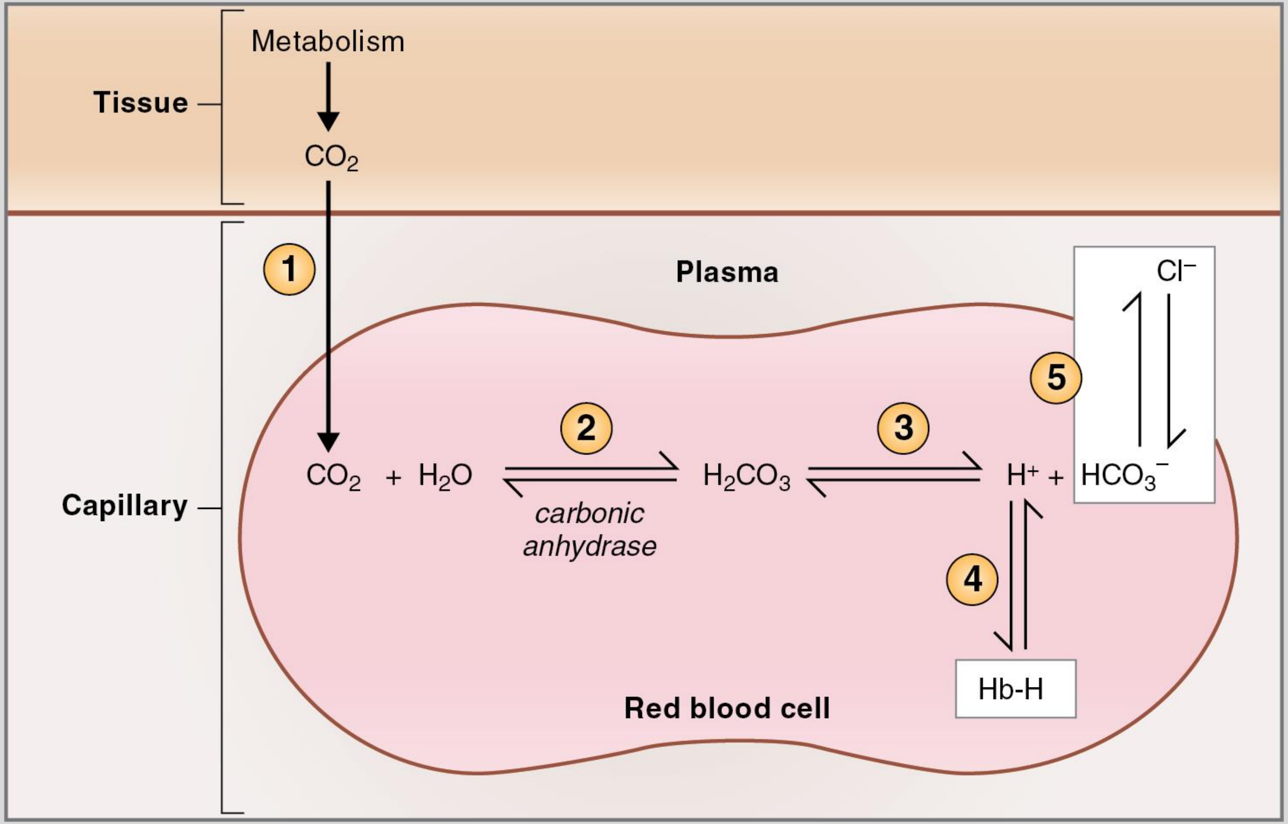

- **Chloride shift** via band-3 protein used for anion exchange
- Deoxy-Hgb: better buffer for H^+

The reverse reactions happen in lung tissue where CO_2 and H_2O are released!

Medical Board-Style Practice Question

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A 19-year-old college track athlete is competing in a high-intensity 400-meter sprint. Midway through the race, his actively contracting skeletal muscle cells experience a sharp increase in metabolic rate, rapidly producing large quantities of carbon dioxide (CO_2) and lactic acid, while generating local heat (hyperthermia). Which of the following best describes the physiological shift occurring in the hemoglobin-oxygen dissociation curve within this localized muscle tissue, and what is its functional benefit?

A. A leftward shift, which increases hemoglobin's affinity for oxygen to ensure the muscle tissue absorbs gas faster.
B. A rightward shift, which decreases hemoglobin's affinity for oxygen, facilitating its unloading to the oxygen-starved muscle.
C. A leftward shift, which stabilizes the relaxed (R) state of hemoglobin to prevent the premature release of oxygen.
D. A rightward shift, which binds oxygen more tightly to hemoglobin to protect the red blood cells from acidic denaturation.
E. No shift occurs, because systemic baroreceptors automatically adjust cardiac output to keep local tissue curves completely constant.

Resources

Updated resources can be found in the Class Preparation Guide (CPG) that goes with this lecture.

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